

University of Southampton
Integrated Polysilicon Photonics
Southampton, United Kingdom

June 17, 2026

To Whom It May Concern

As a passionate AI and Data Science Engineer with a strong background in physics and mathematics, I am thrilled to apply for the PhD position in Artificial Intelligence at the University of Southampton. This opportunity to develop low-cost polysilicon photonic platforms for optoelectronic integration resonates deeply with my research interests and goals.

My previous work on *MedicalMambaTS*, a linear-time state space model for sub-millisecond bedside inference, has allowed me to appreciate the significance of achieving low latency in various cost-laser materials processing procedures to fabricate high-quality polysilicon photonic platforms is a fascinating

Furthermore, my experience with PI-DeepONet, a 0D Windkessel Physics-Informed Operator Network using PyTorch Autograd, has taught me the importance of incorporating physical constraints into machine learning models. I believe that this knowledge will be invaluable in developing low-cost polysilicon photonic platforms for optoelectronic integration.

I am confident that my skills and experience make me an ideal candidate for this PhD position. I look forward to contributing to this project and exploring the vast potential of integrated polysilicon photonics.

Sincerely